Backend test report

2026-10-13 16:15:16 · **132 passed**, **0 failed**, 132 total

Environment

generated	2026-10-13T16:15:15
python	3.13.1
platform	Windows-11-10.0.26200-SP0
cwd	C:\Users\leoam\Documents\Mostly VS Code\Git Files\MRT-Steady-Unsteady
numpy	2.4.6
scipy	1.17.1
matplotlib	3.10.9
rocketcea	1.2.3
pypropep	installed
json5	0.14.0
app_version	1.6
git_commit	db8fae9
git_branch	dev-branch
git_dirty	yes

All tests

Steady (58 configs)

	test	outcome	time	detail
PASS	00_fuel_mass_convergence_baseline	success	3.0s	
PASS	01_parametric_baseline	success	30.0s	
PASS	02_hotfire_baseline	success	0.2s	
PASS	03_hotfire_alt_diameter_only	success	0.2s	
PASS	04_hotfire_high_chamber_pressure	success	0.2s	
PASS	05_hotfire_low_chamber_pressure	success	0.2s	
PASS	06_hotfire_high_ox_flow	success	0.2s	
PASS	07_hotfire_low_ox_flow	success	0.2s	
PASS	08_hotfire_short_grain	success	0.2s	
PASS	09_hotfire_long_grain	success	0.2s	
PASS	10_hotfire_fail_both_alt_filled	validation_error	0.0s	
PASS	11_hotfire_fail_missing_ox_type	validation_error	0.0s	
PASS	12_convergence_high_apogee	success	2.1s	
PASS	13_convergence_low_apogee	success	2.2s	
PASS	14_convergence_impossible_apogee	success	0.5s	
PASS	15_convergence_off_vertical_launch	success	2.1s	
PASS	16_convergence_high_drag	success	1.9s	
PASS	17_convergence_heavy_dry_mass	success	1.5s	
PASS	18_convergence_light_dry_mass	success	2.2s	
PASS	19_convergence_fail_missing_kinematics	validation_error	0.0s	
PASS	20_parametric_sweep_fuel_length	success	17.5s	
PASS	21_parametric_sweep_ox_flow	success	15.1s	
PASS	22_parametric_sweep_chamber_pressure	success	15.8s	
PASS	23_parametric_sweep_two_vars	success	24.2s	
PASS	24_parametric_single_point	success	3.1s	
PASS	25_parametric_fail_low_gt_high	validation_error	0.0s	

	test	outcome	time	detail
PASS	26_parametric_fail_missing_step_size	validation_error	0.0s	
PASS	27_parametric_fail_zero_step_size	validation_error	0.0s	
PASS	28_units_output_mrt_convergence	success	2.6s	
PASS	29_units_output_mrt_hotfire	success	0.2s	
PASS	30_units_output_mrt_parametric	success	35.0s	
PASS	31_units_output_defaults_to_mrt	success	2.5s	
PASS	32_units_output_imp_is_a_no_op	success	2.3s	
PASS	33_units_input_imperial_equivalent	success	2.5s	
PASS	34_units_input_metric_mixed_bag	success	2.4s	
PASS	35_convergence_tight_tolerance	success	3.2s	
PASS	36_convergence_loose_tolerance	success	1.0s	
PASS	37_convergence_min_port_radius_binds	success	0.6s	
PASS	38_convergence_vertical_launch	success	2.5s	
PASS	39_convergence_steep_launch_angle	success	2.1s	
PASS	40_convergence_very_low_apogee	success	2.5s	
PASS	41_hotfire_high_regression_constant	success	0.2s	
PASS	42_hotfire_low_regression_constant	success	0.2s	
PASS	43_hotfire_dense_fuel	success	0.2s	
PASS	44_hotfire_light_fuel	success	0.2s	
PASS	45_hotfire_tight_initial_port	success	0.2s	
PASS	46_hotfire_wide_initial_port	success	0.2s	
PASS	47_parametric_three_variables	success	26.4s	
PASS	48_parametric_sweep_kinematics_variable	success	14.3s	
PASS	49_parametric_sweep_dry_mass	success	9.0s	
PASS	50_parametric_non_integer_step_count	success	5.5s	
PASS	51_parametric_mixed_units_sweep	success	63.4s	
PASS	52_parametric_two_point_sweep	success	6.4s	
PASS	53_fail_missing_regression_exponent	validation_error	0.0s	
PASS	54_fail_hotfire_neither_alt_filled	validation_error	0.0s	
PASS	55_fail_unknown_simulation_type	validation_error	0.0s	
PASS	56_fail_impossible_grain_geometry	validation_error	0.0s	
PASS	58_parametric_all_variables	success	373.3s	

Unsteady (74 configs)

	test	outcome	time	detail
PASS	00_baseline	success	3.2s	
PASS	01_no_warnings	success	3.2s	
PASS	02_graphs	success	18.5s	
PASS	03_stall_luminor_verbatim	engine_abort	0.7s	
PASS	04_stress_high_regression_rate	engine_abort	1.0s	
PASS	05_stress_small_throat	success	3.2s	
PASS	06_stress_near_burnthrough_grain	success	4.0s	
PASS	07_stress_low_feed_pressure_loss	success	3.8s	
PASS	08_stress_high_injector_area	success	3.4s	
PASS	09_stress_low_ullage	success	3.6s	
PASS	10_stress_short_burn	success	2.5s	
PASS	11_stress_long_burn	success	5.2s	
PASS	12_stress_instant_valve	success	3.2s	
PASS	13_fail_missing_cv3_injector	validation_error	0.1s	

	test	outcome	time	detail
PASS	14_fail_cv4_both_alt_filled	validation_error	0.1s	
PASS	15_fail_cv1_alt_neither	validation_error	0.1s	
PASS	16_fail_inner_gt_outer	validation_error	0.1s	
PASS	17_valve_linear	success	3.7s	
PASS	18_phase4a_burnthrough_in_gas_blowdown	success	4.1s	
PASS	19_tank_via_length_branch	success	3.3s	
PASS	20_chamber_via_diameter_branch	success	4.4s	
PASS	21_fuel_limited_burnout	engine_abort	1.4s	
PASS	22_fast_ignition	success	3.5s	
PASS	23_slow_ignition	success	3.9s	
PASS	24_reaches_landing	success	3.2s	
PASS	25_off_vertical_launch	success	2.8s	
PASS	26_high_altitude_launch	success	4.0s	
PASS	27_warn_short_fuel_grain	success	1.9s	
PASS	28_phase4a_burnthrough_thinner_web	success	3.8s	
PASS	32_phase4_large_chamber_volume	success	3.0s	
PASS	33_nozzle_area_ratio_near_unity	success	4.0s	
PASS	35_nozzle_very_high_area_ratio	success	2.9s	
PASS	36_nozzle_large_throat_low_chamber_pressure	success	4.0s	
PASS	37_nozzle_sea_level_launch	success	3.2s	
PASS	38_nozzle_very_high_launch_site	success	3.8s	
PASS	39_nozzle_tiny_throat_high_pressure	success	3.2s	
PASS	40_injector_near_zero_feed_loss	success	3.6s	
PASS	41_injector_huge_feed_loss	success	2.7s	
PASS	42_injector_feed_loss_near_tank_pressure	engine_abort	1.5s	
PASS	43_injector_very_low_discharge_coefficient	success	4.0s	
PASS	44_injector_four_large_holes	success	4.5s	
PASS	45_injector_many_tiny_holes	success	3.9s	
PASS	46_thrust_floor_disabled	success	3.0s	
PASS	47_thrust_floor_aggressive	success	2.9s	
PASS	48_k_amb_raised	success	3.1s	
PASS	51_cea_probe_low_of_ratio	engine_abort	1.2s	
PASS	52_cea_probe_low_chamber_pressure	success	3.7s	
PASS	53_cea_probe_extreme_pressure_tiny_throat	success	3.7s	
PASS	54_n2o_cold_tank_start	engine_abort	0.4s	
PASS	55_n2o_near_critical_tank	success	3.3s	
PASS	56_n2o_floor_via_deep_blowdown	success	2.7s	
PASS	57_terminal_phase_timeout	engine_abort	0.7s	
PASS	58_terminal_quench_during_ignition	engine_abort	1.2s	
PASS	59_terminal_apogee_abort_probe	engine_abort	2.0s	
PASS	60_terminal_metadata_present_on_success	success	3.1s	
PASS	61_units_imperial_equivalent	success	3.2s	
PASS	62_units_metric_mixed_bag	success	3.2s	
PASS	63_units_temperature_celsius_only	success	3.0s	
PASS	64_units_fail_unknown_unit_string	config_error	0.2s	
PASS	65_units_imperial_alternate_branches	success	3.2s	

	test	outcome	time	detail
PASS	66_solver_tight_tolerances	success	9.4s	
PASS	67_solver_loose_tolerances	success	2.2s	
PASS	68_solver_bdf_method	success	5.0s	
PASS	69_solver_large_max_step	success	9.0s	
PASS	70_warn_tight_fuel_port	success	6.2s	
PASS	71_warn_high_ullage	success	3.5s	
PASS	72_warn_low_regression_exponent	success	1.7s	
PASS	73_warn_high_regression_exponent	engine_abort	1.4s	
PASS	74_warn_several_at_once	success	2.0s	
PASS	75_trajectory_vertical_launch	success	3.5s	
PASS	76_trajectory_steep_off_vertical	success	3.1s	
PASS	77_trajectory_very_high_drag	success	2.9s	
PASS	78_trajectory_low_drag_light_vehicle	success	4.2s	
PASS	79_trajectory_tiny_drogue_fast_descent	success	3.5s	

Timing outliers

Far slower than the median for their kind. Worth a look even when they pass: a validation test that takes ten seconds to not reject something never ran its validator.

test	elapsed	median for kind
01_parametric_baseline	30.0s	1.9s
20_parametric_sweep_fuel_length	17.5s	1.9s
22_parametric_sweep_chamber_pressure	15.8s	1.9s
23_parametric_sweep_two_vars	24.2s	1.9s
30_units_output_mrt_parametric	35.0s	1.9s
47_parametric_three_variables	26.4s	1.9s
51_parametric_mixed_units_sweep	63.4s	1.9s
58_parametric_all_variables	373.3s	1.9s